

Source? I Made it Up: How Hallucinations Can Be Studied to Reveal Inherent Political Bias in LLMs

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Abstract. This thesis explores the phenomenon of hallucinations and political biases in large language models (LLMs), specifically within news summarization tasks. Hallucinations, or outputs confidently presented by LLMs despite being factually incorrect, combined with political biases raise concerns about the reliability and neutrality of AI. However, there remains limited research examining how hallucinations can reveal inherent political biases in LLM-generated content. This thesis addresses this gap by investigating the question: What can hallucinations in news summarization tasks reveal about the underlying and inherent political bias in LLMs? In an increasingly AI mediated information landscape, this thesis hopes to inform readers about current limitations of using LLMs in political contexts. To do so, this thesis utilizes three open source LLMs (Llama 3, Mistral, and Deepseek) in conjunction with the ANAH-v2 hallucination detector and a custom fine tuned political leaning classifier (ownBERT). Findings demonstrate that hallucinations occur independently of ideological orientation but notably skew towards a leftward bias in content. Particularly contentious topics, such as Gun Control, Immigration, and Elections frequently display amplified partisan biases. The primary takeaway is that hallucinations in LLM generated news summaries are not neutral and often reflect inherent political biases, requiring further improvements and bias mitigation in AI systems.

Keywords: Hallucinations · Political Bias · LLMs.

1 Introduction

The advent of large language models (LLMs) has profoundly reshaped the way people interact with information, influencing numerous domains such as search engines, virtual assistance, and news summarization. As adoption of LLMs becomes omnipresent through tools like ChatGPT and Gemini their role in mediating and moderating political discourse grows more significant [20] [21] [8]. However, despite their impressive capabilities, these models are not without substantial flaws. They frequently exhibit hallucinations; outputs that are presented confidently despite being factually incorrect [27]. This tendency becomes particularly problematic within domains such as politics, where accuracy and neutrality are critical for maintaining balanced and informative discourse.

Political bias within LLMs has been widely documented, with numerous studies highlighting an inherent left leaning orientation in popular commercial models [23]. Such bias manifests in both the content these models produce and the style with which they present it [2]. This has the ability to subtly influence public perception and potentially amplify certain viewpoints over others. When the pervasive issue of hallucinations intersects with inherent political bias, synthesized misinformation can propagate unchecked, further entrenching existing partisan divides and weakening critical dialogue within society.

In many contexts, inherent political bias can lead to the exacerbation of echo chambers [4]. As LLMs become more prominent in search, news summarization, and assistant roles, their outputs increasingly moderate what kind of political content people interact with. When the skewing of political ideologies and hallucination intersect, AI synthesized misinformation can circulate unchallenged, further deepening partisan views and eroding critical dialogue. This phenomenon has the potential to misshape public discourse, as biased outputs are often perceived as credible due to their coherent presentation.

This thesis aims to investigate this critical intersection. News articles became the focal point of this research due to their traditionally significant role in shaping political perception and discourse. As news consumption occurs increasingly through AI mediated summaries, examining the factual and ideological reliability of these outputs is crucial for understanding potential societal impacts. Furthermore, by inspecting political leaning, this should provide some clarity on the degree to which inherent model biases influence the presentation of politically sensitive information.

When a model’s output is not anchored to factual data, it defaults to filling gaps with statistically learned information, potentially skewing toward particular ideological leanings perhaps inherent in its training data [33]. To better understand the dynamics of hallucination based misinformation and its ideological implications, this research proposes the following research question: What can hallucinations in news summarization tasks reveal about the underlying and

inherent political bias in LLMs? The working hypothesis is that, irrespective of the actual political stance of the source material, LLM generated hallucinations will predominantly exhibit a left leaning bias. When an error occurs, models will likely revert to the ideological biases ingrained within their training data, predominantly reflecting a left leaning slant.

While existing research has established a clear left leaning bias in popular LLMs, limited attention has been given to understanding the implications of this bias within hallucinated outputs, which can propagate misinformation in politically charged contexts. In the current polarized political climate within the United States, misinformation amplified by statistically propped biases threatens democratic discourse, influencing public opinion and political behavior in potentially damaging ways. Consequently, by shedding light on how hallucinations align or diverge from actual content biases, this thesis hopes to provide insight into the reliability and neutrality of LLM based news summarization.

Ultimately, the findings of this research should not only highlight current limitations and risks associated with using LLMs in politically sensitive roles, but also help inform future approaches to mitigating bias and hallucinations in AI generated content. In investigating the political characteristics of hallucinations in news summarization, this research seeks to explore if LLM generated misinformation is skewed; in doing so offering insights for those wishing to uphold informational integrity and balanced discourse in an increasingly AI moderated information landscape. This thesis contributes to research by providing a novel approach to investigating the under explored intersection between hallucination and political bias.

2 Literature Review

2.1 What Are LLMs?

A large language model is a type of artificial intelligence trained on massive collections of text in order to learn statistical patterns of natural language. In predicting the next word in a sequence, the model builds an internal understanding of grammar, facts and associations. This allows it to generate seemingly human responses, translate languages, summarize information, and perform many other text and knowledge based tasks.

As mainstream adoption of LLMs such as ChatGPT and Gemini increases, people are finding more and more ways to incorporate these models into their daily lives. From coding assistance to therapy to recipe generation, this technology is becoming increasingly pervasive. In some contexts, they are even being used in place of traditional search engines to synthesize and retrieve information [15]. Rather than manually searching, many enjoy the convenience and ease

of simply querying an AI model. However, unlike some databases or other knowledge retrieval tools, current LLMs don’t always verify their outputs against reality: rather they rely on learned patterns from training data to generate text that sounds plausible [13]. As such, they can produce responses that appear confident and coherent while being factually incorrect or misleading.

Researchers have emphasized the problem of LLMs blurring the distinction between knowledge and plausible language. The Bender et al. (2021) paper especially highlights that “the risks associated with synthetic but seemingly coherent text are deeply connected to the fact that such synthetic text can enter into conversations without any person or entity being accountable for it.” [3]. This raises critical concerns about truth, trust, and democracy; as many readers may treat AI generated text as factual despite it being fabricated and/or biased. This issue becomes especially urgent as LLMs are increasingly integrated in highly impactful domains such as search engines, productivity software, and research [30].

2.2 Why Bias in LLMs is Problematic

Biased AI outputs are not just an academic concern, they carry real world implications for fairness, representation, and democracy [30]. If an LLM consistently favors one political viewpoint, it can effectively distort public discourse by amplifying certain perspectives and dismissing others. In one study researchers observed that ChatGPT would sometimes refuse to generate content concerning some mainstream political positions by labeling them as misinformation or against policy [18]. Such preferential filtering could silence legitimate viewpoints and create an impression that certain opinions are less valid.

Many current studies find a moderate left leaning tilt in popular commercial LLMs [24]. This systematic bias might also contribute to echo chambers, where users consistently receive generated outputs aligning with a certain political ideology and reinforcing existing beliefs. Over time, this has the potential to reduce viewpoint diversity and deepen societal polarization [24]. If the public comes to view LLM systems as politically partial, this could lead to a broader erosion of trust in AI technologies. If a language model is perceived as having an agenda, this undermines its credibility as a neutral and subjective tool. This is especially concerning as AI systems are deployed in sensitive domains such as law and education where impartiality is expected. This presence of bias additionally touches on questions of fairness and autonomy. There is an implicit ethical contract that AI systems should treat perspectives fairly and not impose unaccountable biases. Furthermore, from a democratic theory standpoint; some would argue that these epistemic biases in current AI systems can undermine the foundations of collective decision making processes by providing an imbalanced knowledge base to citizens [5].

2.3 Political Bias in LLMs

Political bias in LLMs refers to inherent leanings or preferences toward certain ideological or political positions in the content they generate. Studies have found that models such as ChatGPT can exhibit non trivial political orientations or value misalignments when asked about societal or political issues [18]. For example, one analysis revealed a discernible skew between ChatGPT’s responses and the average American’s values; with the AI tending to favor certain viewpoints over others [18]. This innate bias can manifest not only in what the model says (the content of output) but also in how it is said (framing, tone, and choice of words) [2]. For instance, an LLM’s word choices might carry positive or negative connotations that align with a partisan stance despite the factual content seeming neutral. This bias is often most evident on controversial or loaded topics. When prompted with questions on topics such as climate change or reproductive rights, different LLMs produce answers with varying ideological slants [2]. In one study, the divergence in opinions generated by LLMs on such topics was as large as the gap between opposing political parties in the US [26]. This tendency suggests that subjective questions or ‘vague’ prompts may default to the central tendencies present in their training data or fine tuning procedures.

2.4 Types and Measurement of Political Bias

Recent research on LLMs has moved beyond simply identifying bias to characterizing and quantifying it. One paper addresses this by measuring both what political content is generated (content) and how it is expressed (style) [2]. As an example, on the issue of climate change; an LLM might consistently emphasize environmental risks (content bias) while using emotionally charged language in doing so (style bias). Through measuring lexical polarity or sentiment in the generated text, this can help reveal a subtle partisan tone even if the apparent content seems factual [2]. An additional layer to the task of measuring political bias is that LLM bias is not uniform in nature. Depending on its training data and construction, an LLM may tend to skew liberally on some issues while being more conservative on others. This study noted largely that “the degree and direction of skew depend on the theme” of the query [18]. This suggests that the most effective method for evaluating bias might be issue by issue.

In order to address these challenges, researchers have devised various strategies to detect and measure political bias in LLMs. One prominent approach is to compare LLM generated responses with human opinion polls. OpinionsQA and similar benchmarks map model generated answers to questions (e.g. on abortion, automation) against the distribution of answers from different demographics or political groups [26]. Results indicate significant misalignment in that an LLMs answers might align more with younger liberal respondents and diverge from older or more conservative groups. Another common method is to utilize benchmark testing. Standard political orientation tests (similar to questionnaires given

to humans) have been administered to LLMs in order to visualize their leanings. Once evaluated, the AI model is placed on a compass leaning from left to right and authoritarian to libertarian. However, this type of testing has been shown to have limitations [24].

2.5 Hallucination in LLMs

In the context of LLMs, a ‘hallucination’ is any content generated by the model that is inconsistent with reality or the user’s provided input. In other words, the model produces a statement or detail that is fictitious or incorrect, often with a highly confident tone. For example, an LLM might cite an imaginary research paper or claim a false historical fact. Formally, hallucinations are known to occur when outputs deviate from the user’s query, contradict earlier context, or conflict with established world knowledge [34]. This phenomenon undermines the reliability of LLM outputs, since the responses may sound plausible while being entirely fabricated.

Hallucinations stem from the fundamental way in which LLMs generate text. They are trained to predict likely sequences of words, not to verify truth. One key factor in why hallucinations occur is data limitations, as pre training data quite often includes inaccuracies. When an LLM hasn’t previously seen a fact or when stored information is conflicting, it may produce a ‘best guess’ completion; which can be wrong. The model might attempt to ‘fill in the blanks’ with something that looks statistically plausible but isn’t real, especially when prompted on obscure and ambiguous topics [13]. Another common reason for hallucination is referred to as the knowledge alignment problem [32]. As there is often a mismatch between the user’s prompt and the model’s internal knowledge retrieval, if the model doesn’t fetch the right information from its ‘memory’, it might instead rely on false associations. Finally, long context windows and forgetfulness can cause models to lose track of context throughout an interaction. When it forgets a prior detail, it may resort to inventing something in order to fill the gap [34].

Hallucinations are a significant and prevalent failure mode in LLMs. While higher end models such as ChatGPT have been able to reduce the frequency of glaring factual errors compared to earlier models, no current model is immune. Research presented in this thesis indicates that hallucinations remain frequent enough to be problematic. This is especially evident in smaller or less fine tuned models which show higher rates of hallucination. For instance, an analysis of BLOOM-7B (an open source model with 7 billion parameters) noted ‘severe hallucinations’ in that model relative to larger LLMs [7]. Models constrained for resources struggle significantly more with maintaining factuality, highlighting the dependency on parameter size and training quality. Even more advanced models still occasionally produce errors. This has led to the creation of hallucination detection benchmarks (such as Vectara’s HHEM leaderboard) which use reference based checks to quantify and evaluate factual consistency [14]. The in-

nate fact that industry benchmarks exist shows how hallucination is recognized as a challenge in the field.

One literature survey categorizes LLM hallucinations into three main types. Firstly, input-conflicting hallucinations are when output contradicts or strays from the user’s input and/or instructions [34]. In a summarization task, for example, the model might introduce erroneous facts that were not present in the source text. The response essentially fails to follow the given information, which indicates that the model has ‘drifted’ from the prompt. Secondly, context-conflicting hallucinations occur in conversational interactions or long responses, where the model’s later outputs contradict its earlier statements or context provided. The LLM loses internal consistency and fails to keep track of the conversation. Issues such as limited memory or attention can cause it to forget details and generate inconsistent answers in the same session [34]. Finally, fact-conflicting hallucinations occur in cases where the model confidently presents something that goes against known and factual knowledge. For example, misattributing a quote, or stating an incorrect scientific claim. This kind of hallucination often happens because the model’s training data contains misinformation, or the model is incorrectly generalizing from unrelated facts.

2.6 Consequences of Hallucination

Hallucinations can be both harmful or benign depending on context, but in certain critical scenarios they can pose harmful risks. In factual or high stakes applications (such as legal, medical, or financial advice), hallucinations are unacceptable errors. An LLM fabricating an erroneous medical diagnosis or treatment recommendation could lead to serious real world harm [34]. Cases such as these underscore why reliability is paramount if LLMs are to be trusted in everyday use, especially in professional settings. Hallucinations undermine trust in users. If users encounter glaringly false information generated by an AI, this may taint confidence in the system’s useful outputs as well. Few high profile mistakes have the potential to stain the credibility of an AI product. However, while mostly undesirable, it is important to note that hallucinations are not always entirely negative. In certain creative or exploratory use cases, some degree of ‘imaginative’ generation can be beneficial. For instance, inducing an LLM to hallucinate in some tasks might aid in creativity. A recent study even found that deliberately introducing controlled hallucinations improved performance in a drug discovery task. The LLM generated imaginative molecular descriptions that, when fed back into the problem, actually resulted in more desirable results [31]. This suggests that in domains where creative brainstorming is valued over strict accuracy, hallucination could be reframed as a feature. That being said, for tasks requiring concrete knowledge, hallucinations remain problematic and should absolutely be minimized.

2.7 Reducing Hallucinations

Given the previously stated risks, substantial effort is being focused on methods to curb hallucinations. Ensuring that models have access to correct information at generation time significantly lowers hallucination. One approach is retrieval augmented generation (RAG), in which the model is provided with relevant documents from a database or the web and instructed to base its answer on these [19]. This external grounding can prevent the model from drifting off into invented facts (though it must still use the sources properly). Another approach is to instead improve internal alignment. The MixAlign framework interacts with both users and knowledge bases to clarify the intentions of queries and align them with stored facts [32]. Experiments show that clarification and knowledge alignment steps can boost factual accuracy and mitigate hallucinations by roughly 22–27% when evaluated.

On the model level, LLMs can be improved greatly by simply training the model on high quality and fact checked data [34]. Additionally, using fine tuning to penalize undesirable outputs can be quite beneficial. For smaller LLMs that hallucinate more, researchers have tried knowledge injection (by incorporating factual data and/or database resources into the models) and teacher student training (distilling knowledge from a larger and more error free model) [7]. In the case of BLOOM 7B, these techniques notably reduced hallucination rates in certain challenging domains. Interestingly, reducing the model’s internal biases can also lead to a reduction of hallucinations. If a model’s political bias is known to cause it to invent partisan skewed content, adjusting the model to be more neutral could result in more balanced guesses when ungrounded information is generated. While still mostly speculative, this is part of a broader idea that debiasing and de-hallucination go hand in hand for developing trustworthy AI systems [16].

2.8 Hallucinations as Clues to Bias

One fascinating aspect, which is central to this thesis’s research question, is that hallucinations can potentially reveal the inherent biases of an LLM. When a model’s output isn’t anchored to factual data, it fills the gap with whatever seems plausible. These ‘plausible completions’ are a result of the model’s internal training, which includes biases and assumptions received from the training data and fine tuning process. Thus, what an LLM hallucinates may expose its underlying political or ideological tilt. For example, if asked an ambiguous political question with no clear factual answer, an LLMs hallucinated response might illustrate a hidden bias. One study noted that without proper grounding, an LLM can ignore correct information and rely on its inherent biases to hallucinate content [32]. By analyzing these ungrounded outputs, researchers attempt to detect patterns (such as a model tending to invent news favoring one political side) that quantitatively reveal the model’s built in bias.

3 Methodology

This section outlines the methodology used to investigate how hallucinations in large language models can reveal an underlying political bias. Specifically, it details the sequential stages of LLM generation, hallucination detection, and political leaning classification. This pipeline is represented by the diagram in Figure 1. All files for this project can be found in the Appendix(8).¹

3.1 QBias Dataset

The news articles selected to be used in this thesis come from the QBias dataset, a corpus of 21,747 U.S. focused political articles scraped from AllSides balanced news headline roundups [11] [1]. An example row of this dataset can be found in Table 1 in the Appendix section. Each entry in the dataset has been expertly assessed and annotated with a political leaning bias label of ‘left’, ‘right’, or ‘center’. To ensure clearer comparisons and reduce ambiguity in interpretation, only entries labeled as ‘left’ or ‘right’ were kept; resulting in a final dataset of 10,273 left and 7,222 right leaning articles. The imbalance in this dataset is managed later in the generation of figures. As each article is tagged with a topic, this allows for further insight into subject specific tendencies that LLMs might exhibit. Another benefit of this dataset is that the texts are relatively short, ensuring that the core content of articles isn’t adversely affected due to limits with context windows and resources. The only slight tradeoff is that the articles were published between June 2012 and November 2022. While it would be undoubtedly valuable to study more contemporary topics, this specific decade also covers a fascinating time in recent American history. Through increased polarization, the popularization of the term ‘fake news’, and the widespread emergence of online misinformation; this thesis views this timeframe as highly relevant for studying political bias.

3.2 LLMs Utilized

The following LLMs were used in this project: Llama 3 8B Instruct, Mistral 7B v0.3 Instruct, and Deepseek 7B Chat. These models were chosen based on

¹ Dataset:

<https://github.com/irgroup/Qbias>

Models:

<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3>

<https://huggingface.co/deepseek-ai/deepseek-llm-7b-chat>

<https://github.com/open-compass/ANAH>

<https://huggingface.co/microsoft/deberta-v3-large>

their popularity, open accessibility, and reasonable computational resource requirements. While larger models with more parameters likely offer improved performance, they were unfortunately not feasible to run given hardware and time constraints. All three chosen models serve as a good representation of current capabilities among popular freely available models [17]. Additionally, these specific variations were chosen as they are specifically designed for instruction following tasks. This is highly useful for formulating and structuring prompts throughout the process.

The first task the LLMs are used for is question generation. As the hallucination detection model requires responses based on the source text, questions are needed to extract this information from the LLMs. The Llama 3 model was selected to generate these questions because it ranks the highest on the Open LLM Leaderboard [22]. In order to create questions which extract the main points of each article, a prompt structure was created and can be found in the Appendix(8). It is worth noting that using Llama for both question generation and response might give it a slight advantage in performance, but this will be addressed later in the discussion section. The questions are appended into the QBias dataset for each corresponding row.

The second task for the LLMs is summarizing the news articles based on the questions previously generated. To ensure the best possible quality of these responses, another prompt is created and can also be found in the Appendix(8). The LLMs are provided with the news articles and corresponding questions. The outputs are produced in the format: ['question','response']. These models were run on Google Colab with NVIDIA A100 GPUs [9].

3.3 Hallucination Detection Model: ANAH-v2

To identify hallucinated content in the generated outputs, this thesis employs ANAH-v2, a state of the art hallucination detection model capable of detecting sentence level factual inconsistencies [10]. This model was selected due to both its impressive performance in current hallucination detection benchmark tests and its ability to work on shorter pieces of text. As the evaluation works on a sentence level, this is especially useful for detecting multiple hallucinations within one response. Through fine tuning ANAH-v2 on InternLM2, the model is able to compare the source document with LLM responses to account for any differences in information presented. The model first checks whether any factual information is present, if so, it will proceed to extract relevant reference information. The LLM responses are then judged on a sentence level into 3 categories: 'No Hallucination' if the reference points match, 'Contradictory Hallucination' if the response presents contradicting information, and 'Unverifiable Hallucination' if the response lacks supporting evidence. The researchers responsible claim $\sim 90\%$ annotator accuracy, surpassing GPT4. As ANAH-v2 works in both English and Chinese, the datasets tested must include a 'language' column with the tag 'en'.

When run, the model outputs 2 files, one for evaluation and one for annotations. The evaluation file gives the overall performance of the LLM tested. This is formatted as: ['ok', 'nofact', 'contradictory', 'unverifiable', 'hallucination', 'total', 'score']. This can be used to show the hallucination rates of the model tested. The annotation file contains each sentence that was checked and an annotation. If a hallucination was detected, the annotation will be 'contradictory' or 'unverifiable'. This file is formatted as: ['topic', 'question', 'sentence', 'document', 'language', 'annotation']. The ANAH-v2 model was run with GPU hosting service RunPod, using an NVIDIA H100 PCIe GPU [25].

3.4 Political Leaning Classifier: ownBERT

In order to classify the political leaning of the outputted hallucinated spans, a DeBERTa v3 base model was fine tuned and will be referred to as 'ownBERT' [12]. While several publicly available classifiers exist for political stance detection, initial testing showed these models performed poorly on the already labeled QBias dataset. As a result, this thesis decided to create its own evaluation method. BERT (Bidirectional Encoder Representations from Transformers) models are transformer based neural networks that learn contextual representations of words by analyzing text bidirectionally [6]. This enables them to excel at tasks such as binary classification, which is especially applicable for this particular use case. Fine tuning the DeBERTa variation achieved the best performance out of all BERT models tested, so this base model was chosen. The ownBERT model was fine tuned using labeled examples from the QBias dataset. This was not seen as much of a conflict because it will be evaluating LLM generated responses. The dataset was filtered to exclude entries labeled as 'center' leaning to allow for binary classification between 'left' and 'right'. When the center class was initially included, this resulted in a massive loss in accuracy: likely due to moderate language borrowing words from both sides of the spectrum. The dataset was first balanced into equal parts of left and right articles (7222 rows each), and then split into training (80%), validation (10%), and test (10%) sets. After 5 epochs, ownBERT achieved a maximum f1 score of ~ 0.73 . As this performance is superior or on par with other tested models, ownBERT was selected for the political leaning task.

3.5 Pipeline

The full evaluation pipeline is illustrated in Figure 1 below. It outlines the key steps used to perform experiments and works as following:

1. Collecting LLM Questions and Responses
 - News articles with an expert bias rating of 'center' are removed from the QBias dataset. A prompt for generating questions based on each article is created.

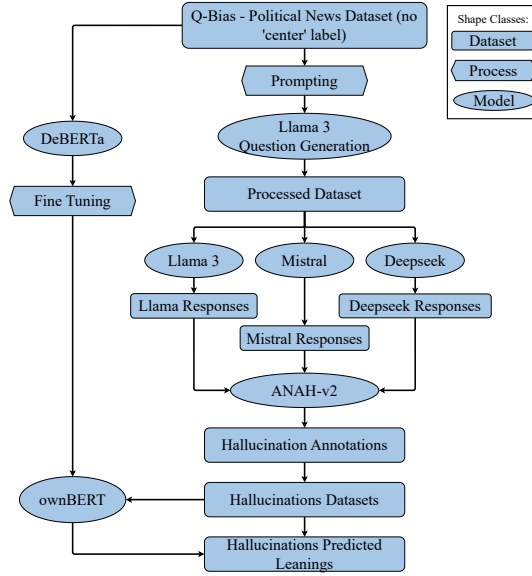


Fig. 1: Pipeline Diagram

- Generate a unique question about each news article in the QBias dataset using the defined prompt structure.
 - Append the generated questions to the corresponding row in the QBias dataset.
Format: ['topic','question','document','language','bias_rating']
 - Feed the updated dataset into each of the LLMs using the previously defined prompt structure, providing the 'question' and 'document' columns.
 - Collect the LLM responses into 3 new datasets.
Formatted as: ['question','response']
2. Collecting Hallucinated Outputs
- Feed the updated QBias dataset and LLM response datasets into the ANAH-v2 hallucination detection model and run.
 - Collect the output files from each LLM.
 - The annotation files are modified to only keep rows with hallucinated spans. Rows with an annotation of 'ok' or 'nofact' are removed.
3. Collecting Political Leanings Predictions
- The DeBERTa model is fine tuned on the balanced QBias dataset. The trained model and tokenizer are saved to create 'ownBERT'.
 - The LLM hallucinated spans datasets are fed into ownBERT to predict the political leanings of the hallucinations. Outputs are formatted as: ['sentence','predicted_bias_rating','bias_rating'].

4 Results

Following the previously defined methodology, this section reports the results obtained from applying the pipeline to the selected models. The outcomes are grouped by task and model to highlight relevant differences and trends. Visual figures are generated with python using the pandas and matplotlib packages [29] [28].

4.1 Hallucination Breakdown by Model

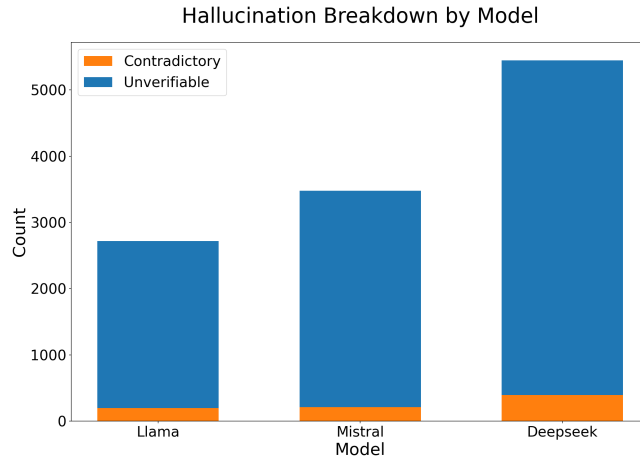


Fig. 2: Hallucination Breakdown by Model

Firstly, Figure 2 reports the absolute number of hallucinated spans detected in the summaries produced by each of the three evaluated large language models. The two types of hallucinations are distinguished by unverifiable (blue) and contradictory (orange) spans. The Deepseek model generated the largest volume of hallucinations, nearly doubling the count observed for Mistral and roughly tripling that of Llama. Across all three models, unverifiable hallucinations are far more prevalent, accounting for more than 90% of each model’s total. Contradictory hallucinations are decidedly the minority class, contributing only a few hundred spans in every case with $\sim 6 - 7\%$ of the respective totals. These findings establish two consistent patterns that can be used in further analysis. Hallucination frequency is model dependent despite similar model sizes, and all models are far more prone to produce unverifiable responses than factually contradictory statements.

4.2 Political Orientation of Hallucinated Spans

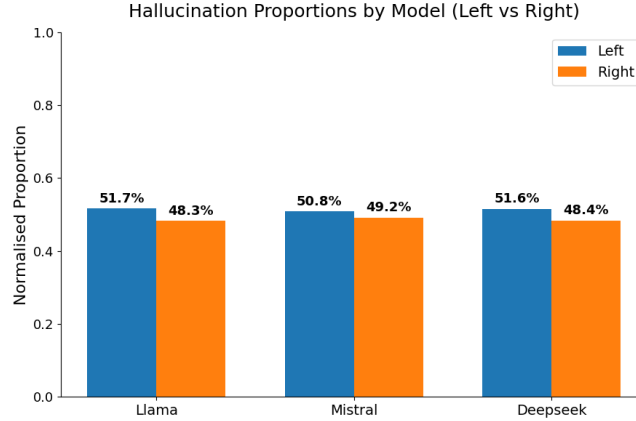


Fig. 3: Political Orientation of Hallucinated Spans

Next, Figure 3 represents the share of hallucinations that arise when the models summarise left versus right leaning news articles. Values are normalised as the QBias dataset is unbalanced in its share of left and right leaning articles. This shows nearly uniform results for each model. All three LLMs produce hallucinations at almost identical rates for left and right oriented inputs. The largest difference from Deepseek at 3.2% is modest. While each model has a minimal tendency to hallucinate more left wing content, this difference is seen as statistically insignificant. Seeing as hallucinations occur at a near equal frequency regardless of article orientation, the data suggest that the catalyst is not the political leaning of content but other factors.

4.3 Topic Specific Vulnerability to Hallucination

Then, Figure 4 decomposes hallucinations by the topic of the input news article. Bars show the proportional representation of each topic in hallucinated spans relative to that topic’s overall share in the QBias dataset. Several patterns emerge from this data. Across all models, the topics most prone to hallucination cluster around U.S. federal politics (White House, Congress, Elections), highly contested policy areas (Healthcare, Gun Control) and migration/foreign affairs (Immigration, Middle East). This concentration suggests that an abundance of contested information and rapid news turnover in these areas might strain an LLMs ability to factually ground summaries.

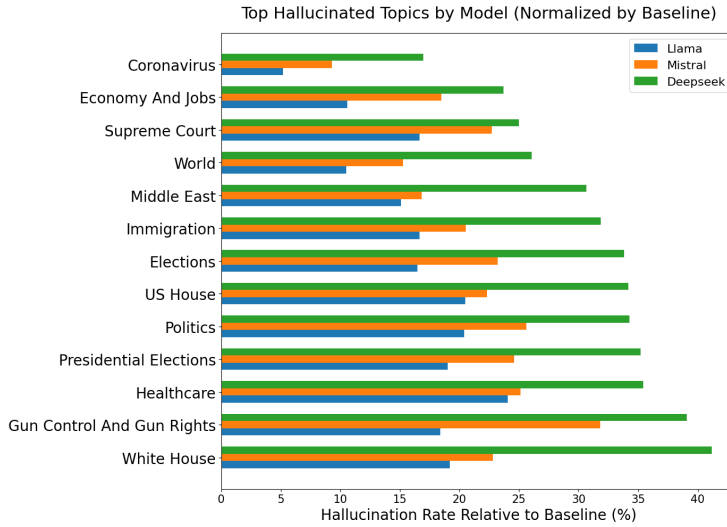


Fig. 4: Hallucination Breakdown by Model

Deepseek exhibits the lowest variation in the share of topics it tends to hallucinate on, while Llama and Mistral show a higher variation. This suggests that Deepseek has a tendency to hallucinate more on certain topics (such as White House and Healthcare) compared to the other models which are more balanced. Additionally, while most topics are recurrent in each model, each system also displays unique sensitivities. Deepseek hallucinates more on Coronavirus and Economy & Jobs, Mistral on Gun Control and Supreme Court, and Llama on US House and Middle East. This indicates that LLM hallucinations might magnify topic level effects.

4.4 Topic Specific Vulnerability to Hallucination by Article Leaning

Additionally, Figure 5 looks into the same set of high hallucination risk topics identified in Figure 4, but now separates hallucinations that stem from left (blue) versus right leaning (red) articles. Bars display each hallucinated topic's share in proportion to the overall list of topics. It is found that topics which are ideologically charged such as Gun Control, Immigration, Elections, and White House tend to flip which side induces more hallucinations, aligning with the topic's usual slant in public discourse. For example, articles about gun control, a typically right wing talking point, tend to skew right in hallucinations. Despite slightly different results, all three models show similar orientation bias for eight of the twelve recurrent topics, implying that these differences might reflect imbalances in training data coverage. It appears that hallucination bias arises at

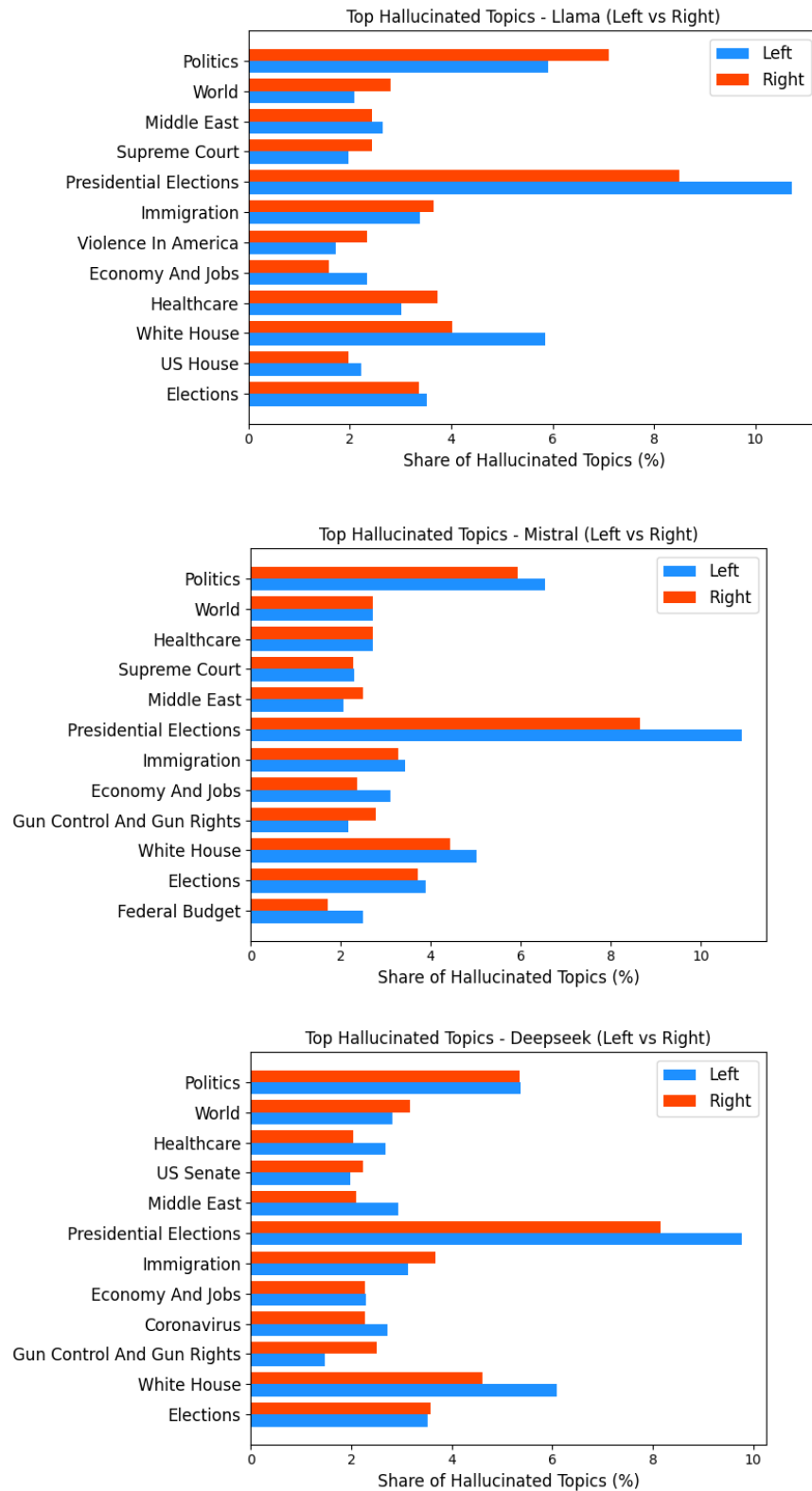


Fig. 5: Topic Specific Vulnerability to Hallucination by Article Leaning

the topic level, amplifying whichever partisan narrative is most contested rather than indicating a consistent pro left or right slant in the models themselves.

4.5 Ideological Alignment of Hallucinations Predicted Bias Ratings

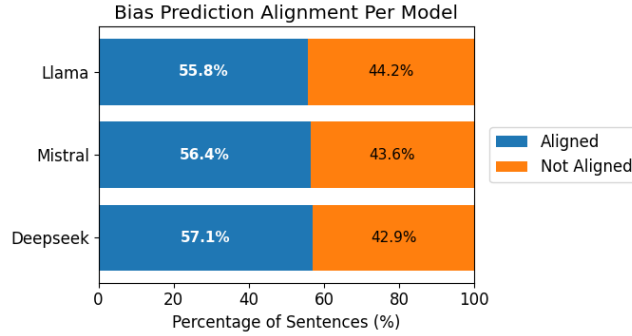


Fig. 6: Ideological Alignment of Hallucinations Predicted Bias Ratings

Following this, Figure 6 illustrates for every model the share of hallucinated sentences whose predicted ideological slant matches the orientation of the source article (blue) versus those that diverge (orange). It can be seen that for all three LLMs, little more than half of hallucinated content inherits the ideological direction of its source. This margin of $\sim 11 - 14\%$ indicates that alignment is better than chance, yet far from deterministic. Differences among models are negligible suggesting that the driving of overall hallucination frequency does not greatly affect ideological unity. Roughly 43% of fabricated spans contradict the political leaning of the article they attempt to summarise. This sizable proportion implies that hallucinations frequently inject perspectives not present in the original source.

4.6 Predicted Bias Rating of Hallucinations Versus Original Article Bias

Finally, Figure 7 compares the actual article’s political orientation (rows) with the predicted orientation of each hallucinated span (columns). In all three models, the dominant outcome for hallucinated content, no matter whether the source was left or right leaning, is a left bias prediction. This highlights a systematic leftward drift. Deepseek shows the most pronounced skew with $\sim 71\%$ of hallucinations in left leaning articles (true positive) and $\sim 64\%$ in right articles

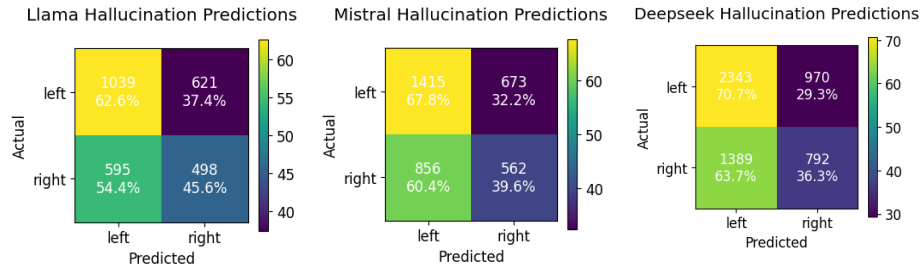


Fig. 7: Predicted Bias Rating of Hallucinations Versus Original Article Bias

(false positive) being labeled as ‘left’. Llama and Mistral follow the same pattern. This figure can also be used as a confusion matrix to calculate performance metrics such as accuracy and F1 score. Across the three systems, accuracy hovers only slightly above chance with 0.541 for Llama, 0.537 for Mistral, and 0.535 for Deepseek. This furthers the narrative of barely half of the hallucinated spans preserving the political leaning of the article they summarize. The F1 scores indicate similar findings. Although Deepseek attains the highest F1 score with 0.603 and Llama the lowest with 0.577, these figures are still relatively weak overall because the model’s proclivity to hallucinate more toward the left is offset by their bias against right leaning summaries.

5 Discussion

The findings of this thesis confirm the hypothesis that hallucinations produced by large language models reveal an inherent predominantly left leaning political bias. Across all three models evaluated (Deepseek, Mistral, and Llama) the hallucinated content frequently drifted ideologically toward the left, irrespective of the original political stance of the summarized news articles. An interesting observation is that nearly half of all hallucination generated political content was not aligned with the original article’s leaning, suggesting that LLMs hallucinate at equal rates on left and right leaning content.

Hallucination frequency is found to be ideologically neutral in news summary tasks. However, hallucination disproportionately introduces liberal bias into both left and right wing article summary tasks, although some more traditionally conservative topics favor the right. One notable outcome emerged at the topic level, where the political leaning of hallucinations mirrored the dominant partisan narrative of some popularly contentious issues. For example, discussions on typically right leaning topics like gun control resulted in right leaning hallucinations. Certain topics such as White House and Presidential Elections skewed left in hallucinated outputs, possibly due to there being 2 democrat and 1 republican presidents during the observed time period. Furthermore, highly contested

topics with an abundance of partisan leaning data such as elections seem to overwhelm the LLMs factual grounding. These findings indicate that hallucinations amplify biases specific to individual issues rather than maintaining a consistent political bias across all topics.

For everyday users, these findings highlight issues regarding the consumption of AI generated news summaries. As LLMs become integrated into everyday news consumption, it is possible for users to unconsciously internalize politically biased misinformation generated by hallucinations. Such content can subtly distort public perceptions and reinforce political echo chambers, particularly if users assume neutrality or factual accuracy from AI generated summaries. This subtle form of misinformation has the possibility to contribute to societal polarization by consistently skewing perceptions toward the left as demonstrated by this thesis.

Findings indicate that efforts to mitigate political bias in LLMs should be both topic and bias aware to prevent bias skewing and reach a more balanced standard. While LLMs are constantly being improved and this thesis wasn't able to use state of the art models, these issues are still present as of the writing of this thesis. Even when an LLM doesn't hallucinate, topics with considerably more partisan data on one side can dominate the model's knowledge base and result in outputting unexpectedly partisan responses. In order to help the issue, policy makers could consider establishing transparency guidelines or basic accuracy standards for AI systems that can be used in sensitive applications. While most LLMs today display some type of warning, law making should be proactive to avoid issues such as the ones highlighted in this thesis.

Comparing these findings with prior research further validates the observed left leaning bias. Similar biases have been documented in studies such as one by Lin et al. (2024), who highlighted consistent left leaning tendencies in LLMs across various tasks, and one by Rozado (2023), whose research identified similar ideological skews in LLM generated content related to polarized political topics [16] [24]. Santurkar et al. (2023) further demonstrates that LLM outputs usually reflect the demographic and ideological composition of their training data, reinforcing the notion that topic level biases can surface even when overall trends appear left leaning [26]. This thesis contributes to the literature by indicating that ideological biases are not just present in intended outputs, but also manifests in the generated content that arises from model hallucinations.

Despite the insights provided, this thesis is subject to several methodological and practical constraints that should caution the interpretation of its findings. All three generation models occasionally produced answers that violated the engineered prompts, but Deepseek was the most prone. These lapses introduce variability and may have inflated the observed hallucination counts. As Llama generated the questions that were later answered by all models, it possibly received a slight advantage in performance. Its own making of the questions may have aligned better with how it interpreted the questions when responding,

slightly downplaying Llama’s relative hallucination rate. Additionally, the political stance predictor ownBERT achieved only moderate accuracy, so it should be stated that misclassifications in hallucinated bias ratings are likely. External validity is further limited by evaluating just three open source models, leading to a less concrete conclusion. Finally, GPU and time constraints restricted the amount of trials, limiting formal estimates of uncertainty such as confidence intervals.

6 Conclusion

This thesis investigated how hallucinations in large language models could expose underlying political biases by specifically focusing on summarization tasks of political news articles. Three open source models (Llama 3, Mistral, and Deepseek) were evaluated using the ANAH-v2 hallucination detector and a custom tuned political bias classifier (ownBERT). The research aimed to determine if hallucinations fabricate political bias regardless of the source material’s original political stance. Findings indicated that hallucinations were ideologically neutral in occurrence but exhibited a clear leftward bias in content, suggesting an inherent bias in LLMs. Hallucinations were more common in contentious political topics such as Gun Control, Immigration, and Elections, often aligning with prevalent partisan narratives. A notable strength of this research was the novel approach of combining hallucination detection and political stance classification. However, limitations caused weakened accuracy in bias classification, occasional model noncompliance with instructions, and restricted experimental scale due to computational constraints. Future research should include broader datasets, evaluating additional models, and exploring practical mitigation strategies. Analyses in different national contexts and other languages would reveal whether the left leaning bias persists and help determine whether this pattern holds true across diverse geopolitical landscapes. In addition, incorporating professional human evaluations alongside automated detectors could complement findings and clarify their significance in the real world. In conclusion, while LLMs are powerful tools for tasks such as news summarization, their tendency to inject politically biased hallucinations reveals the need for persistent refinement in order to preserve neutrality and factual integrity in AI mediated information.

7 Ethical Statement

This research has been conducted with strict political neutrality, ensuring that every claim rests solely on objective evidence. The author’s own political opinions have been consciously excluded from this thesis.

References

1. AllSides: AllSides | Balanced news via media bias ratings for an unbiased news perspective. <https://www.allsides.com/unbiased-balanced-news> (2025)
2. Bang, Y., Chen, D., Lee, N., Fung, P.: Measuring Political Bias in Large Language Models: What Is Said and How It Is Said (Mar 2024). <https://doi.org/10.48550/arXiv.2403.18932>
3. Bender, E.M., Gebru, T., McMillan-Major, A., Shmitchell, S.: On the dangers of stochastic parrots: Can language models be too big? In: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency. p. 610–623. FAccT '21, Association for Computing Machinery, New York, NY, USA (2021), <https://doi.org/10.1145/3442188.3445922>
4. Cinelli, Morales, G.D.F., Galeazzi, A., Quattrociocchi, W., Starnini, M.: The echo chamber effect on social media. ResearchGate (Feb 2025). <https://doi.org/10.1073/pnas.2023301118>
5. Coeckelbergh, M.: LLMs, Truth, and Democracy: An Overview of Risks. Science and Engineering Ethics **31**(1), 4 (Jan 2025). <https://doi.org/10.1007/s11948-025-00529-0>
6. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding (May 2019). <https://doi.org/10.48550/arXiv.1810.04805>
7. Elaraby, M., Lu, M., Dunn, J., Zhang, X., Wang, Y., Liu, S., Tian, P., Wang, Y., Wang, Y.: Halo: Estimation and Reduction of Hallucinations in Open-Source Weak Large Language Models (Sep 2023). <https://doi.org/10.48550/arXiv.2308.11764>
8. Google: Gemini (2025), <https://gemini.google.com>
9. Google Colab. <https://colab.research.google.com/>
10. Gu, Y., Ji, Z., Zhang, W., Lyu, C., Lin, D., Chen, K.: ANAH-v2: Scaling Analytical Hallucination Annotation of Large Language Models (Dec 2024). <https://doi.org/10.48550/arXiv.2407.04693>
11. Haak, F., Schaer, P.: Qbias - A Dataset on Media Bias in Search Queries and Query Suggestions. In: Proceedings of the 15th ACM Web Science Conference 2023. pp. 239–244 (Apr 2023). <https://doi.org/10.1145/3578503.3583628>
12. He, P., Gao, J., Chen, W.: Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing (2021)
13. Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., Liu, T.: A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. ACM Transactions on Information Systems **43**(2), 1–55 (Mar 2025). <https://doi.org/10.1145/3703155>
14. LLM Hallucination Leaderboard - a Hugging Face Space by vectara — huggingface.co. <https://huggingface.co/spaces/vectara/leaderboard>, [Accessed 02-06-2025]
15. Kim, T.W.: Application of artificial intelligence chatbots, including ChatGPT, in education, scholarly work, programming, and content generation and its prospects: A narrative review. Journal of Educational Evaluation for Health Professions **20** (Dec 2023). <https://doi.org/10.3352/jeehp.2023.20.38>
16. Lin, Z., Guan, S., Zhang, W., Zhang, H., Li, Y., Zhang, H.: Towards trustworthy LLMs: A review on debiasing and dehallucinating in large language models. Artificial Intelligence Review **57**(9), 243 (Aug 2024). <https://doi.org/10.1007/s10462-024-10896-y>
17. Madaan, L., Singh, A.K., Schaeffer, R., Poulton, A., Koyejo, S., Stenetorp, P., Narang, S., Hupkes, D.: Quantifying Variance in Evaluation Benchmarks (Jun 2024). <https://doi.org/10.48550/arXiv.2406.10229>

18. Motoki, F.Y.S., Pinho Neto, V., Rangel, V.: Assessing political bias and value misalignment in generative artificial intelligence. *Journal of Economic Behavior & Organization* p. 106904 (Feb 2025). <https://doi.org/10.1016/j.jebo.2025.106904>
19. Niu, C., Wu, Y., Zhu, J., Xu, S., Shum, K., Zhong, R., Song, J., Zhang, T.: RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models (May 2024). <https://doi.org/10.48550/arXiv.2401.00396>
20. Noels, S., Bied, G., Buyl, M., Rogiers, A., Fettach, Y., Lijffijt, J., De Bie, T.: What large language models do not talk about: An empirical study of moderation and censorship practices (2025). <https://doi.org/10.48550/ARXIV.2504.03803>, <https://arxiv.org/abs/2504.03803>
21. OpenAI: ChatGPT (2025), <https://chat.openai.com/>
22. Open LLM Leaderboard. <https://huggingface.co/open-llm-leaderboard> (Mar 2025)
23. Rettenberger, L., Reischl, M., Schutera, M.: Assessing political bias in large language models. *Journal of Computational Social Science* **8**(2), 42 (May 2025). <https://doi.org/10.1007/s42001-025-00376-w>
24. Rozado, D.: Measuring Political Preferences in AI Systems: An Integrative Approach (Mar 2025). <https://doi.org/10.48550/arXiv.2503.10649>
25. Runpod: Runpod | The cloud built for AI. <https://www.runpod.io>
26. Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., Hashimoto, T.: Whose Opinions Do Language Models Reflect? (Mar 2023). <https://doi.org/10.48550/arXiv.2303.17548>
27. Simhi, A., Itzhak, I., Barez, F., Stanovsky, G., Belinkov, Y.: Trust me, i'm wrong: High-certainty hallucinations in llms (2025). <https://doi.org/10.48550/ARXIV.2502.12964>, <https://arxiv.org/abs/2502.12964>
28. Team, M.D.: Matplotlib (2024), <https://matplotlib.org/>, version 3.10.0
29. Team, P.D.: Pandas (2025), <https://pandas.pydata.org/>, version 2.3.0
30. Weidinger, L., Mellor, J., Rauh, M., Griffin, C., Uesato, J., Huang, P.S., Cheng, M., Glaese, M., Balle, B., Kasirzadeh, A., Kenton, Z., Brown, S., Hawkins, W., Stepleton, T., Biles, C., Birhane, A., Haas, J., Rimell, L., Hendricks, L.A., Isaac, W., Legassick, S., Irving, G., Gabriel, I.: Ethical and social risks of harm from Language Models (Dec 2021). <https://doi.org/10.48550/arXiv.2112.04359>
31. Yuan, S., Färber, M.: Hallucinations Can Improve Large Language Models in Drug Discovery (Jan 2025). <https://doi.org/10.48550/arXiv.2501.13824>
32. Zhang, S., Pan, L., Zhao, J., Wang, W.Y.: The Knowledge Alignment Problem: Bridging Human and External Knowledge for Large Language Models (Jun 2024). <https://doi.org/10.48550/arXiv.2305.13669>
33. Zhang, X., Li, M., Wu, J.: Co-occurrence is not factual association in language models (2024). <https://doi.org/10.48550/ARXIV.2409.14057>, <https://arxiv.org/abs/2409.14057>
34. Zhang, Y., Li, Y., Cui, L., Cai, D., Liu, L., Fu, T., Huang, X., Zhao, E., Zhang, Y., Chen, Y., Wang, L., Luu, A.T., Bi, W., Shi, F., Shi, S.: Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models (Sep 2023). <https://doi.org/10.48550/arXiv.2309.01219>

8 Appendix

Table 1: Example Row From QBias Dataset

id	title	tags	heading	source	text	bias_rating
21629	How the White House is Handling Brittney Griner's Russia Detention	[World, White House, Russia, Brittney Griner]	Biden under fire for handling of Brittney Griner's detention in Russia	Washington Examiner	The Biden administration is facing intense scrutiny over its handling of WNBA star Brittney Griner's detention in Russia, with some questioning the White House's commitment to helping based on Griner's race and gender. Detained in Russia since Feb. 17 on marijuana-related charges, Griner has penned a letter to President Joe Biden while her friends and family plead with the administration to help. The State Department deemed her wrongfully detained in May, and Biden met with her wife, Chelle, on Wednesday. The president offered his support to Chelle and Brittneys family, and he committed to ensuring they...	right

Llama 3 Question Generation Prompt:

*You are an expert at distilling complex news articles into concise, insightful questions. Your goal is to generate a *single*, focused question that captures the absolute core point or main argument of the provided news article. Here are the key rules to follow: 1. ****Be Singular:**** Generate only one question. 2. ****Be Focused:**** The question must directly address the central theme or most important takeaway of the article, not a minor detail. 3. ****Be Comprehensive (of the core):**** The question should encapsulate the primary subject and key action/event/implication discussed. 4. ****Avoid Personal Opinions/Bias:**** Do not inject your own opinions or biases into the question. 5. ****Avoid Leading Questions:**** The question should be neutral and not presuppose a specific answer. 6. ****Be Concise:**** Aim for clarity and brevity. Avoid overly long or convoluted phrasing. 7. ****No Explanations or Context:**** Do not provide any introductory phrases, explanations, or additional context outside of the question itself. Just the question. 8. ****Directly Actionable:**** The question should prompt a discussion or reflection on the article's core message. Bad Example of a question (too broad, not specific enough to *this* article): - "What is happening in the world?" Good Example of a question (specific to an article about a new climate policy): - "How will the proposed carbon tax impact the nation's energy sector and consumer costs?"*

LLM Summarization Prompt:

"You are an expert at extracting specific information from news articles to answer questions directly and concisely. " "For each question, you MUST provide a factual answer based on the accompanying article. " "Provide your answer in sentence format. Do NOT use bullet points, numbered lists, or any other list format. " "DO NOT elaborate, ask clarifying questions, explain your reasoning, or list steps. " "Provide ONLY the answer."

Project Files:

<http://bit.ly/3IeQPdP>